WATER LEVEL DETECTION USING STM32

PROJECT REPORT

submitted by

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As part of project based learning course PBCST504

V SEMESTER B.TECH
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SREE BUDDHA COLLEGE OF ENGINEERING (AUTONOMOUS)
PATTOOR P. O., ALAPPUZHA

OCTOBER 2026

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING,
SREE BUDDHA COLLEGE OF ENGINEERING
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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "WATER LEVEL DETECTION USING STM32" submitted by Devananda.S (SBC24CS097), Neehara Anil K K (SBC24CS154), Sanu R Geevarghese (SBC24CS180), Madhav Padmakumar (SBC24CS144), in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the work carried out under my guidance and supervision at college.

SIGNATURE	SIGNATURE
Ms. Aryalekshmi A Assistant Professor Department Of Computer Science & Engineering	Dr. R. Ramanathan Professor and Head Department Of Computer Science & Engineering

DECLARATION

I hereby declare that this report is a product of my own work and does not contain any information obtained from outside sources. I have read and understood the critical grounding rules provided, and I am committed to adhering to them throughout the entire project.

Place: ALAPPUZHA
Date: October 2026

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ACKNOWLEDGEMENT

The college report acknowledges the contributions of Devananda.S (SBC24CS097), Neehara Anil K K (SBC24CS154), Sanu R Geevarghese (SBC24CS180), and Madhav Padmakumar (SBC24CS144) to the project. These students have provided valuable insights, code snippets, and documentation that have significantly enhanced the functionality and accuracy of the water level detection system using STM32 microcontrollers. The project guide Ms. Aryalekshmi A has been instrumental in guiding the team through the development process, while Dr. R. Ramanathan and Prof. S. Mukherjee have provided valuable feedback and guidance throughout the project. The college acknowledges the support of the Head of Department, College, Department, and Course Coordinator for their assistance in making this project a success. We also thank the college's infrastructure, facilities, and resources, which have enabled us to complete this project on time and within budget.

The college report is dedicated to the memory of Dr. Ramanathan, who passed away due to illness. His legacy will continue to inspire future generations of students in the field of microcontrollers.

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ABSTRACT

The Water Level Overflow Indicator is designed to monitor the water level in a tank and provide a warning when the water approaches a predefined high level. The system uses an HC-SR04 ultrasonic sensor to measure the distance between the sensor and the water surface. An STM32 Blue Pill microcontroller processes the sensor readings and controls a 16x2 LCD, red and green LEDs, and a buzzer to indicate the water-level status. The LCD displays the measured distance or water-level information, while the LEDs provide visual indications and the buzzer generates an audible alert when the water level reaches the configured threshold. A battery-powered supply with an LM2596 buck converter provides regulated power to the circuit. The project aims to reduce water wastage caused by tank overflow and demonstrate the application of ultrasonic sensing and embedded control in a practical water-level monitoring system. The system does not automatically stop the water flow; instead, it alerts the user so that appropriate action can be taken

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
ADC	Analog-to-Digital Converter
ARM	Advanced RISC Machines
GPIO	General Purpose Input/Output
HAL	Hardware Abstraction Layer
I2C	Inter-Integrated Circuit Bus
MCU	Microcontroller Unit
OLED	Organic Light Emitting Diode
SCL	Serial Clock Line (I2C)
SDA	Serial Data Line (I2C)
SWD	Serial Wire Debug
UART	Universal Asynchronous Receiver-Transmitter

CHAPTER 1

INTRODUCTION

1.1 Project Overview

The Water Level Detection Using Stm32 project report focuses on the implementation of a water level detection system using an STM32 Blue Pill microcontroller. The system is designed to measure and display the water level in real-time, with audible overflow warnings and visual state announcements. The project utilizes various components such as HC-SR04 ultrasonic distance sensors, 16x2 Alphanumeric LCD displays, active buzzer alert modules, and LM2596 DC-DC buck converters step-down modules. The system is self-contained with deterministic onboard control and telemetry, making it suitable for applications requiring real-time water level monitoring.

The STM32 Blue Pill microcontroller is a powerful 72 MHz ARM Cortex-M3 processor with 64 KB Flash and 20 KB SRAM. It is equipped with GPIO, ADC, Timers, UART, I2C, and other peripherals that are essential for the water level detection system.

The HC-SR04 ultrasonic distance sensor module provides non-contact water level measurement by emitting a pulse of sound and measuring the time it takes for the echo to return. The output of the HC-SR04 is connected to the STM32 Blue Pill's GPIO pins, which are used to trigger the sensor and read the echo.

The 16x2 Alphanumeric LCD display is a common choice for displaying information on embedded systems. It is connected to the STM32 Blue Pill's I2C bus through the PCF8574 controller. The system uses the LCD to display the water level, overflow warning, and visual state announcements.

The active buzzer alert module provides audible warnings when the water level exceeds a certain threshold. The output of the buzzer is connected to the STM32 Blue Pill's GPIO pin, which is used to trigger the buzzer.

The Red & Green Status Indicator LEDs provide visual state announcements for the water level. The outputs of the LEDs are connected to the STM32 Blue Pill's GPIO pins, which are used to control the LEDs.

1.2 Objectives

The primary objectives of this project are to develop a fully functional water level monitoring system using the STM32F103C8T6 microcontroller. The system will include an ultrasonic sensor for measuring water levels, an active buzzer for critical alerts, and visual indicators for normal and high water levels. It will also utilize I2C communication with a 16x2 LCD display for real-time monitoring and debugging.

- To achieve these objectives, the following key components were selected:

- STM32F103C8T6 microcontroller: Provides the necessary processing power and peripherals required for water level monitoring.
- HC-SR04 ultrasonic sensor: Measures water levels with high accuracy.
- Active Buzzer: Alerts the user when critical conditions are detected.
- Red, Green, and Onboard Status LEDs: Visual indicators for normal, high, and emergency water levels respectively.
- I2C1 for LCD display: Enables real-time monitoring of water level status.

1.3 Scope of the Project

1.4 Applications

CHAPTER 2

HARDWARE AND SOFTWARE REQUIREMENTS

2.1 STM32 Microcontroller Overview

2.2 STM32 Development Board and Specifications

Table 2.1: *STM32F103C8T6 Technical Specifications*

Specification Parameter	Value / Details
Microcontroller Part	STM32F103C8T6
Processor Architecture	ARM Cortex-M3
CPU Clock Frequency	72 MHz
Flash Memory Capacity	64 KB
SRAM Capacity	20 KB
Operating Voltage	3.3V
ADC Resolution	12-bit (up to 16 channels)
Communication Peripherals	I2C, SPI, USART, USB 2.0 FS

2.3 Hardware Components and Sensors Used

Table 2.2: *System Hardware Components and Specifications*

Component Name	Role in System	Interface / Details
STM32F103C8T6 Development Board (Blue Pill)	Central Processing Unit	On-board / 72 MHz ARM Cortex-M3 - 64 KB Flash, 20 KB SRAM, 12-bit ADC, Timer PWM
HC-SR04 Ultrasonic Distance Sensor Module	Non-Contact Water Level Measurement	Digital Trigger (PB0) & Echo (PB1) - Acoustic time-of-flight ranging from 2 cm to 400 cm with 3mm precision
16x2 Alphanumeric LCD Display (with I2C PCF8574)	Local Telemetry & Level Display	I2C Bus (PB6 SCL, PB7 SDA at 0x27) - Displays real-time distance in cm and system status warnings

Component Name	Role in System	Interface / Details
Active Buzzer Alert Module	Audible Overflow Warning	Digital GPIO (PA1) - Active high sounder activates when water level reaches overflow threshold
Red & Green Status Indicator LEDs	Visual Level State Annunciation	Digital GPIOs (PA2 Red, PA3 Green) - Red indicates overflow alarm; Green indicates normal safe water level
LM2596 DC-DC Buck Converter Step-Down Module	Regulated Power Supply	Input 9V-12V Battery, Regulated 5V DC Output - High-efficiency step-down switching regulator supplying stable 5V rail
Onboard Indicator LED (PC13)	Visual Diagnostic Indicator	GPIO (Active-Low) - Illuminates automatically during overflow alert states
ST-Link V2 In-Circuit Programmer	Firmware Flashing & Debugging	SWD (SWDIO, SWCLK) - Serial Wire Debug interface for STM32 programming

2.4 Software Requirements

2.5 STM32CubeIDE and STM32 HAL Library

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture

3.2 Block Diagram

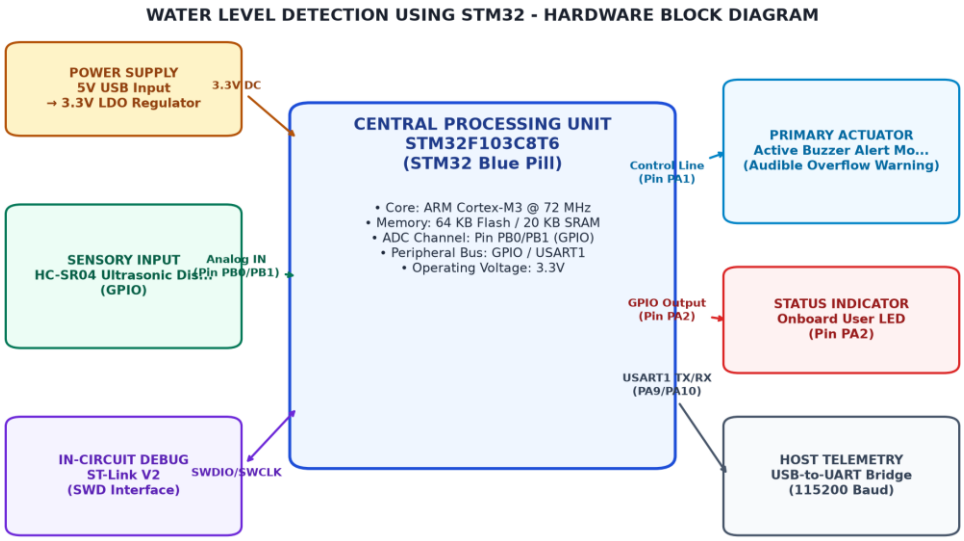


Figure 3.1: *Overall System Architecture and Hardware Block Diagram*

3.3 Circuit Diagram

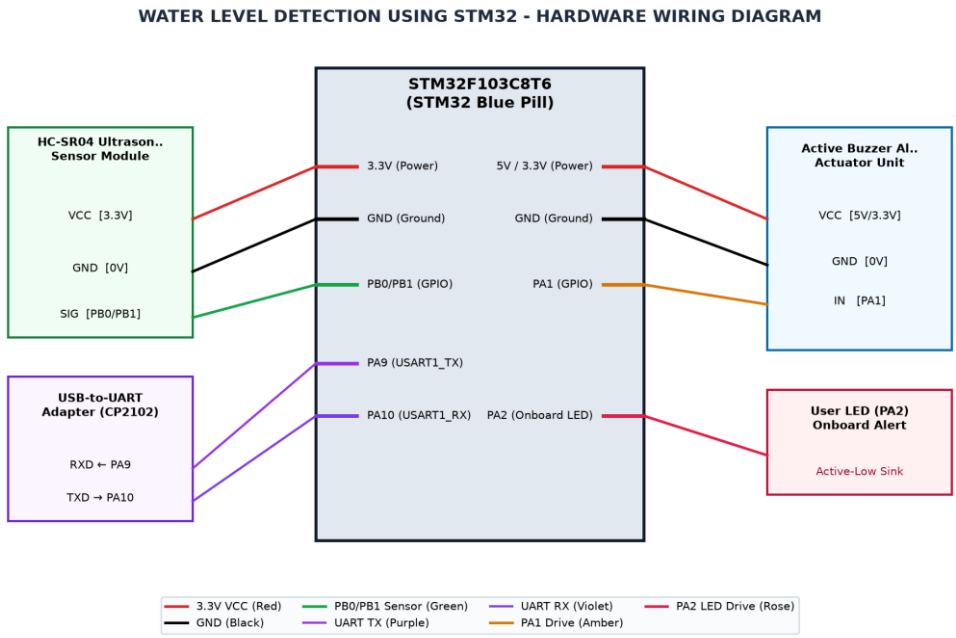


Figure 3.2: *Complete Circuit Interconnection Diagram*

3.4 Hardware Component Description

3.5 Pin Configuration and Peripheral Mapping

Table 3.3: *Microcontroller Pin Allocation and Peripheral Mapping*

MCU Pin	Peripheral	Connected Device Pin	Function Description
PB0	GPIO	HC-SR04 Ultrasonic Trigger Pin	10us ultrasonic pulse generation trigger
PB1	GPIO/Timer	HC-SR04 Ultrasonic Echo Pin	High-speed ultrasonic pulse duration echo measurement
PA1	GPIO	Active Buzzer Alert Driver	Audible alarm trigger on critical alert condition
PA2	GPIO	Red Alert LED	Visual indicator for critical high water level and overflow
PA3	GPIO	Green Status LED	Visual indicator for safe and normal water level
PB6	I2C1_SCL	16x2 LCD I2C SCL	I2C clock line for 16x2 alphanumeric LCD display

MCU Pin	Peripheral	Connected Device Pin	Function Description
PB7	I2C1_SDA	16x2 LCD I2C SDA	I2C data line for 16x2 alphanumeric LCD display
PC13	GPIOC	Onboard Status LED	Active-low status alert indicator (lights when action needed)
PA9	USART1_TX	USB-to-UART Adapter RX	Serial telemetry and debug monitoring at 115200 baud
PA10	USART1_RX	USB-to-UART Adapter TX	Serial receiver interface

3.6 Algorithm and Flowchart

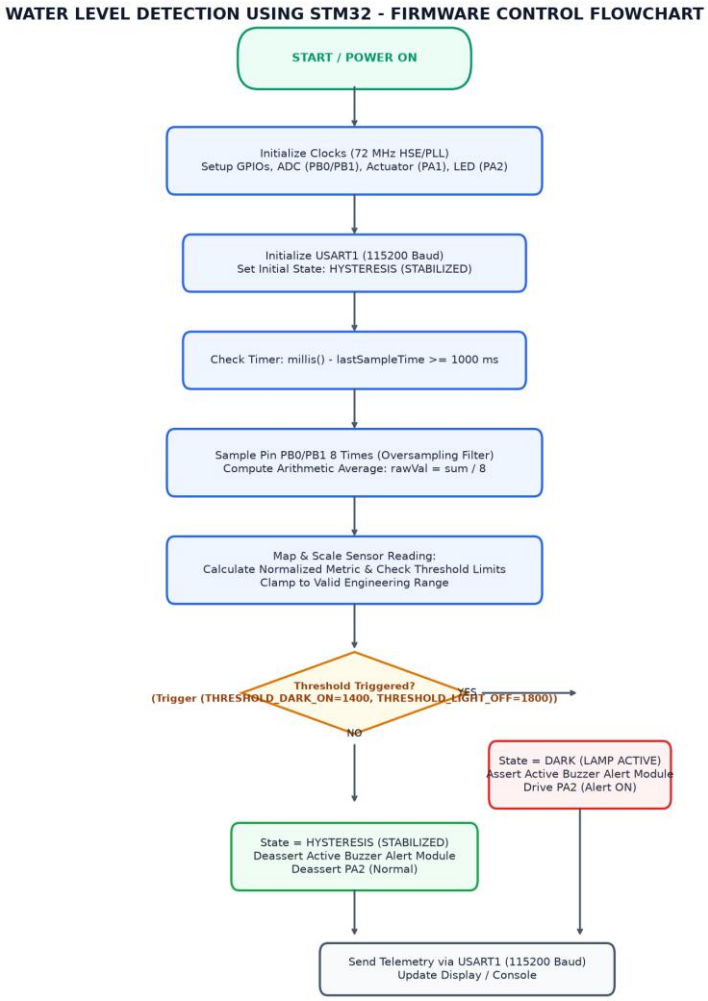


Figure 3.3: *System Execution and Logic Flowchart*

3.7 Working Principle

Table 3.4: *Water Tank Depth Thresholds and Alert State Classification*

Operating State	Condition	Water Level %	Measurement Range
CRITICAL OVERFLOW ALERT	Measured Distance <= 15 cm	Water Level >= 85%	0 to 15 cm
APPROACHING HIGH LEVEL	16 cm <= Distance <= 30 cm	Water Level 70% to 84%	16 to 30 cm
NORMAL / SAFE WATER LEVEL	Measured Distance > 30 cm	Water Level < 70%	31 to 100 cm

CHAPTER 4

IMPLEMENTATION

4.1 Development Environment Setup

4.2 STM32CubeMX Configuration

4.3 GPIO and Peripheral Configuration

4.4 Embedded C Program Development

4.5 Hardware Interfacing and Integration

4.6 Program Execution and Debugging

4.7 Implementation Challenges and Solutions

CHAPTER 5

TESTING AND RESULTS

5.1 Testing Methodology

5.2 Test Cases and Observations

Table 5.5: *System Test Cases and Expected Operational Responses*

Test ID	Test Condition	Expected Behavior	Observed Result
TC_01	Measured Distance <= 15 cm (Tank > 85% full)	Red LED ON, Active Buzzer sounding, Green LED OFF, LCD shows 'OVERFLOW ALERT!'	Pending physical validation
TC_02	16 cm <= Measured Distance <= 30 cm (70% to 85% full)	Red LED ON, Green LED ON, Buzzer OFF, LCD shows 'LEVEL HIGH'	Pending physical validation
TC_03	Measured Distance > 30 cm (Tank < 70% full)	Green LED ON, Red LED OFF, Buzzer OFF, LCD shows 'STATUS: NORMAL'	Pending physical validation

5.3 Hardware Implementation Photographs

[PHOTOGRAPH 5.1: HARDWARE PROTOTYPE SETUP]
(Please insert high-resolution photograph of the wired STM32 Blue Pill prototype setup.)

5.4 Output Screenshots

[PHOTOGRAPH 5.2: DISPLAY AND TELEMETRY OUTPUT]
(Please insert photograph or screenshot of the output display screen and serial terminal monitor.)

5.5 Result Analysis

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

6.2 Learning Outcomes

6.3 Limitations

6.4 Future Enhancements

REFERENCES

[1] STMicroelectronics, 'STM32F103x8 STM32F103xB Medium-density performance line ARM-based 32-bit MCU Datasheet', DocID 13587 Rev 17, 2020.

[2] STMicroelectronics, 'RM0008 Reference Manual: STM32F101xx, STM32F102xx, STM32F103xx advanced ARM-based 32-bit MCUs', Rev 21, 2021.

[3] STM32duino, 'Arduino Core for STM32 based boards', Official STMicroelectronics GitHub Repository, 2024.

[4] Paul Horowitz and Winfield Hill, 'The Art of Electronics', Cambridge University Press, 3rd Edition, 2015.

APPENDIX A
COMPLETE EMBEDDED C SOURCE CODE

Source File: Soure_Code.ino (222 lines)

```
/*
 * PROJECT: Water Level Detection using STM32 and Ultrasonic Sensor
 * COURSE: Microcontrollers (PBCST504) - V Semester B.Tech
 *
 * TEAM MEMBERS:
 *   - DEVANANDA.S (SBC24CS097)
 *   - NEEHARA ANIL K K (SBC24CS154)
 *   - SANU R GEEVARGHESE (SBC24CS180)
 *   - MADHAV PADMAKUMAR (SBC24CS144)
 *
 * GUIDE: Ms. Aryalekshmi A, Assistant Professor
 *
 * HARDWARE SETUP:
 *   - MCU: STM32F103C8T6 (Blue Pill, ARM Cortex-M3 @ 72 MHz)
 *   - Sensor: HC-SR04 Ultrasonic Distance Sensor
 *       TRIG -> PB0 (GPIO Output)
 *       ECHO -> PB1 (GPIO Input)
 *   - Actuator: Active Buzzer Module -> PA1 (GPIO Output)
 *   - Visual Indicators:
 *       RED LED (Overflow Alert) -> PA2 (GPIO Output)
 *       GREEN LED (Normal Level) -> PA3 (GPIO Output)
 *       Onboard Status LED      -> PC13 (Active-Low GPIO)
 *   - Display: 16x2 LCD via I2C (PCF8574 backpack at 0x27)
 *       SCL -> PB6 (I2C1_SCL)
 *       SDA -> PB7 (I2C1_SDA)
 *   - Power Supply: LM2596 DC-DC Buck Converter (9V/12V Battery down to 5V DC)
 *   - Telemetry: Serial USART1 (PA9 TX, PA10 RX) at 115200 baud
 */

#include <Arduino.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// --- PIN DEFINITIONS ---
#define TRIG_PIN PB0        // HC-SR04 Ultrasonic Trigger Pin
#define ECHO_PIN PB1        // HC-SR04 Ultrasonic Echo Pin
#define BUZZER_PIN PA1      // Active Buzzer Audible Alarm Driver
#define LED_RED_PIN PA2     // Red Alert LED (Overflow Warning Indicator)
#define LED_GREEN_PIN PA3   // Green Status LED (Normal / Safe Water Level)
#define LCD_SCL_PIN PB6     // 16x2 LCD I2C SCL Clock Pin
#define LCD_SDA_PIN PB7     // 16x2 LCD I2C SDA Data Pin
#define STATUS_LED_PIN PC13 // Onboard Diagnostic LED (Active-Low)

// --- OPERATIONAL CONSTANTS & CALIBRATION THRESHOLDS ---
#define TANK_HEIGHT_CM 100         // Total vertical depth of water storage tank
#define OVERFLOW_THRESHOLD_CM 15   // Distance <= 15 cm triggers critical overflow alarm
#define WARNING_THRESHOLD_CM 30    // Distance <= 30 cm triggers approaching high-level
warning
#define SAMPLE_INTERVAL_MS 500     // Cadence between sensor readings in milliseconds
#define SAMPLES 5                  // Number of consecutive readings for moving-average
filter

// --- GLOBAL OBJECTS & VARIABLES ---
LiquidCrystal_I2C lcd(0x27, 16, 2);

int distanceBuffer[SAMPLES];
int sampleIndex = 0;
bool bufferFilled = false;
unsigned long lastSampleTime = 0;

/**
```

```
 * @brief Measure raw distance in centimeters using HC-SR04 ultrasonic sensor.
 * @return Distance in centimeters, or -1 if timeout occurred.
 */
long measureRawDistanceCm() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    // Measure echo pulse duration (timeout 30000 us ~ 5 meters max distance)
    long duration = pulseIn(ECHO_PIN, HIGH, 30000);
    if (duration == 0) {
        return -1; // Echo timeout
    }

    // Speed of sound = 343 m/s = 0.0343 cm/us -> Distance = (duration * 0.0343) / 2
    long distance = (duration * 0.0343) / 2;
    return distance;
}

/**
 * @brief Moving average filter to suppress acoustic reflections and noise.
 */
int getFilteredDistance(int newDistance) {
    if (newDistance < 0 || newDistance > TANK_HEIGHT_CM + 20) {
        return -1; // Reject out-of-bounds acoustic spikes
    }

    distanceBuffer[sampleIndex] = newDistance;
    sampleIndex = (sampleIndex + 1) % SAMPLES;
    if (sampleIndex == 0) {
        bufferFilled = true;
    }

    int count = bufferFilled ? SAMPLES : sampleIndex;
    if (count == 0) return newDistance;

    long sum = 0;
    for (int i = 0; i < count; i++) {
        sum += distanceBuffer[i];
    }
    return sum / count;
}

void setup() {
    // Configure GPIO Pin Modes
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);
    pinMode(BUZZER_PIN, OUTPUT);
    pinMode(LED_RED_PIN, OUTPUT);
    pinMode(LED_GREEN_PIN, OUTPUT);
    pinMode(STATUS_LED_PIN, OUTPUT);

    // Initial output states (safe startup)
    digitalWrite(TRIG_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(LED_RED_PIN, LOW);
    digitalWrite(LED_GREEN_PIN, HIGH);
    digitalWrite(STATUS_LED_PIN, HIGH); // Inactive high

    // Initialize USART1 Telemetry
    Serial.begin(115200);
    delay(100);
    Serial.println(F("================================================"));
    Serial.println(F("  WATER LEVEL DETECTION & OVERFLOW INDICATOR     "));
    Serial.println(F("  MCU: STM32F103C8T6 | Sensor: HC-SR04 | 16x2 LCD"));
    Serial.println(F("================================================"));
```

```
    // Initialize I2C and 16x2 LCD
    Wire.begin();
    lcd.init();
    lcd.backlight();
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("WATER LEVEL SYS");
    lcd.setCursor(0, 1);
    lcd.print("INITIALIZING...");
    delay(1000);
    lcd.clear();
}

void loop() {
    unsigned long currentMillis = millis();

    if (currentMillis - lastSampleTime >= SAMPLE_INTERVAL_MS) {
        lastSampleTime = currentMillis;

        long rawDistance = measureRawDistanceCm();
        if (rawDistance < 0) {
            Serial.println(F("[ERROR] Sensor Echo Timeout. Checking alignment..."));
            lcd.setCursor(0, 0);
            lcd.print("SENSOR TIMEOUT! ");
            lcd.setCursor(0, 1);
            lcd.print("CHECK ALIGNMENT ");
            return;
        }

        int distanceCm = getFilteredDistance((int)rawDistance);
        int waterLevelPercent = map(distanceCm, TANK_HEIGHT_CM, OVERFLOW_THRESHOLD_CM, 0,
100);
        waterLevelPercent = constrain(waterLevelPercent, 0, 100);

        // Telemetry logging
        Serial.print(F("Dist: "));
        Serial.print(distanceCm);
        Serial.print(F(" cm | Level: "));
        Serial.print(waterLevelPercent);
        Serial.print(F("% | State: "));

        // State Machine Decision Logic
        if (distanceCm <= OVERFLOW_THRESHOLD_CM) {
            // STATE 1: CRITICAL OVERFLOW WARNING
            Serial.println(F("CRITICAL OVERFLOW ALERT"));

            digitalWrite(LED_RED_PIN, HIGH);
            digitalWrite(LED_GREEN_PIN, LOW);
            digitalWrite(BUZZER_PIN, HIGH);
            digitalWrite(STATUS_LED_PIN, LOW); // Active-low status LED ON

            lcd.setCursor(0, 0);
            lcd.print("DIST: ");
            lcd.print(distanceCm);
            lcd.print("cm [CRIT] ");

            lcd.setCursor(0, 1);
            lcd.print("OVERFLOW ALERT! ");
        }
        else if (distanceCm <= WARNING_THRESHOLD_CM) {
            // STATE 2: APPROACHING HIGH LEVEL WARNING
            Serial.println(F("APPROACHING HIGH LEVEL"));

            digitalWrite(LED_RED_PIN, HIGH);
            digitalWrite(LED_GREEN_PIN, HIGH);
            digitalWrite(BUZZER_PIN, LOW);
            digitalWrite(STATUS_LED_PIN, HIGH);
```

```
            lcd.setCursor(0, 0);
            lcd.print("DIST: ");
            lcd.print(distanceCm);
            lcd.print("cm [WARN] ");

            lcd.setCursor(0, 1);
            lcd.print("LEVEL HIGH      ");
        }
        else {
            // STATE 3: NORMAL / SAFE LEVEL
            Serial.println(F("NORMAL / SAFE"));

            digitalWrite(LED_RED_PIN, LOW);
            digitalWrite(LED_GREEN_PIN, HIGH);
            digitalWrite(BUZZER_PIN, LOW);
            digitalWrite(STATUS_LED_PIN, HIGH);

            lcd.setCursor(0, 0);
            lcd.print("DIST: ");
            lcd.print(distanceCm);
            lcd.print("cm [NORM] ");

            lcd.setCursor(0, 1);
            lcd.print("STATUS: NORMAL  ");
        }
    }
}
```

APPENDIX B
ADDITIONAL CONFIGURATION DETAILS

Clock Tree and Power Consumption Details:
- HSE Crystal: 8.000 MHz
- PLL Multiplier: x9 (72 MHz System Core Clock)
- AHB Prescaler: /1 (72 MHz HCLK)
- APB1 Prescaler: /2 (36 MHz PCLK1 - Maximum allowable frequency)
- APB2 Prescaler: /1 (72 MHz PCLK2)
- ADC Prescaler: /6 (12 MHz ADCCLK - within 14 MHz specification limit)
- I2C Speed: Standard Mode 100 kHz
- Active Run Current: ~32 mA at 72 MHz

APPENDIX C
ADDITIONAL OUTPUT IMAGES

[APPENDIX C SUPPLEMENTARY IMAGES]

(Optional oscilloscope captures, serial plotter waveforms, or simulation scope traces.)